NOT MEASUREMENT SENSITIVE

MIL-G-45204C
AMENDMENT 3
9 June 1993
SUPERSEDING
AMENDMENT 2
12 March 1984

GOLD PLATING, ELECTRODEPOSITED

This amendment forms a part of Military Specification MIL-G-45204C, dated 7 June 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.2: Add the following ASTM documents:
"ASTM B567 - Measurement of Coating Thickness by the Beta Backscatter Method

ASTM B568 - Measurement of Coating Thickness by X-Ray Spectrometry

ASTM B578 - Microhardness of Electroplated Coatings"

PAGE 3

3.4, line 2: Include 6.3 within the parenthesis so that it now reads: "(see 6.2, 6.3)".

3.4: Add the following sentence to the end of the paragraph: "Unless otherwise specified, gold over silver underplate combinations shall be excluded from electronics hardware. Silver or copper plus silver may not be used as an underplate unless required by item specification. Also, when gold is applied to a copper rich surface such as brass, bronze or beryllium copper or a copper plate or strike, an antidiffusion underplate such as nickel shall be applied".

3.5, line 3: Add "(see 6.2.1)" at the end of the sentence.

PAGE 4

3.6.6: Delete last two sentences.

PAGE 5

4.2: Delete last sentence.

PAGE 6

4.3.2, line 2: Insert after "sample" the phrase "from each lot"

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4.3.2: Add the following sentence: "When separate specimens are used for the hardness test and the as-plated gold composition (purity) test they shall be in accordance with 4.3.2.1, 4.3.2.1.1, 4.3.2.1.1.1, and 4.3.2.1.1.2."

PAGE 6

Add as new paragraphs 4.3.2.1.1.1 and 4.3.2.1.1.2:

"4.3.2.1.1.1 Frequency of hardness test using separate specimens. Hardness testing of deposit (gold plating) shall be determined initially on a weekly basis and should be continued until process stability and control are demonstrated (see 4.5.3). Documented process performance (data) shall be the indicator that the process is in control. Unless otherwise specified, once the hardness test data is in conformance with the hardness range specified for the purity (type) and hardness (grade) designation, the frequency of the test can be revised to allow adjustment based upon process history. Once there is evidence that the process is stable and in control, the frequency of the hardness test shall be performed on a monthly basis, unless there is evidence that the process is no longer in control (see 4.7). If data indicates that the hardness is out of range, the weekly test should be reinstituted until the process is stable and in control.

4.3.2.1.1.2 Frequency of testing versus process control. The hardness of the deposit is directly related to the condition of the bath. Process control procedures and process parameters have been developed by the suppliers of the chemical plating solutions to ensure that the plated products will meet the requirements of the military specification. Process control recommended by these suppliers involve measurement of the gold content, evaluation of the pH, verification of solution density, and analysis of the alloy hardening additives and other metal impurities in the bath. A spectrographic analysis of the electrodeposited gold coating will confirm that the purity of the deposit relates to the composition or purity specified for use (see 3.6.7, 4.5.7).

NOTE: These process controls assure that the process and product remain consistent at each plating facility. With adequate control, the purity and hardness of the plated deposits will be consistent, and testing once per month will be sufficient to assure that the plating will perform as intended."

PAGE 8

4.5.1.2, line 1: Insert after "determined" the phrase "in accordance with ASTM B567".

4.5.1.3, line 1: Insert after "determined" the phrase "in accordance with ASTM B568".

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PAGE 9

4.5.3: Delete and substitute:

"4.5.3 Hardness. The hardness shall be determined on the cross section of the plating in accordance with ASTM B578 and ASTM E384 by the use of the Knoop hardness tester using a test load of 25 gram-force. In accordance with ASTM E384, the Knoop hardness may be determined perpendicular to the plated surface if the thickness of the plate is at least 10 times the depth of the Knoop indentation."

PAGE 10

4.6.1, line 4: Delete "Disposition of Nonconforming Product" and substitute "Acceptance and Rejection."

4.7: Delete and substitute:

"4.7 Reduced testing. At the discretion of the procuring activity, the frequency of testing may be reduced provided the process can be shown to be in statistical control and the manufacturer can ensure product conformance."

6.1.1 Add the following paragraph:

"6.1.1 Soldering. Metal finishers report that gold or gold with codeposited impurities can make soldering more difficult. For this reason high purity gold coatings may be preferred. General purpose and high reliability engineering practices often specify soldering applications. Failure analysis data from space applications indicates that these soldering requirements are best achieved when gold coatings range between 0.00005 and 0.0001 inch in thickness."

PAGE 11

6.2.1: Add the following item:

"(e) Location of contact mark(s). The size and number of contact marks will be minimal and consistent with drawings/good practice. The location of contact mark(s) are usually preagreed areas which will not be exposed to the environment or which will not hamper the function of the part."

6.3, line 7: Delete the second and last sentence because both are inserted in paragraph 3.4.

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